$$= \frac{15+16}{2} = \frac{31}{2} = 15.5$$

Here, to find median we arrange the values of the given data either in ascending or descending order, then find the average of the two middle values.

So we conclude that, to find median,

(i) arrange the data in ascending or descending order.

(ii) If the number of terms (n) is odd, then $\left(\frac{n+1}{2}\right)^{th}$ term is the median.

(iii) If the number of terms (n) is even, then average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ terms is the median.

Try these

1. Find the median of 3, 8, 7, 8, 4, 5, 6.
2. Find the median of 11, 14, 10, 9, 14, 11, 12, 6, 7, 7.

Activity

Create a group of 6 to 7 students and collect the data of weight of the students in your class. In each group find mean, median and mode. Also, compare the averages among groups. Are they same for all the groups?

Also find all the three averages for entire class. Now, compare the results with the average of each of the groups.

Example 5.10 Find the median of the following golf scores.

68, 79, 78, 65, 75, 70, 73.

Solution

Arranging the golf scores in ascending order, we have,

65, 68, 70, 73, 75, 78, 79

Here $n = 7$, which is odd.

Therefore, Median $= \left(\frac{n+1}{2}\right)^{th}$ term

$= \left(\frac{7+1}{2}\right)^{th}$ term.

$= \left(\frac{8}{2}\right)^{th}$ term

$= 4^{th}$ term $= 73$

Hence, the Median is 73.

7th_Maths_Stat_Ch5.indd 106 05-11-2019 17:37:14